

ARACAJU, Brazil

WMO: 830950

Lat: 10.984S

Lon: 37.070W

Elev: 7

StdP: 101.24

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	20.9	21.8	16.9	12.1	25.3	17.8	12.8	25.2	8.8	24.9	8.3	25.3	1.7	310	0.431

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	5.2	32.1	26.6	31.5	26.4	31.0	26.2	27.2	30.7	26.8	30.2	26.6	30.0	5.4	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.2	21.6	29.4	25.9	21.3	29.4	25.3	20.5	28.9	86.0	31.0	84.4	30.6	83.1	30.3	29.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.3	7.7	7.2	19.1	33.2	0.7	2.4	18.5	34.9	18.1	36.3	17.7	37.6	17.2	39.3		
			17.8	27.9	1.4	1.3	16.8	28.8	16.0	29.6	15.2	30.3	14.1	31.2		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	26.7	27.8	28.1	28.1	27.6	26.6	25.8	25.1	24.9	25.7	26.6	27.0	27.5
	DBStd	1.48	0.95	1.10	1.03	1.12	1.04	0.99	1.00	1.04	1.00	1.03	0.86	0.85
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6107	553	507	560	528	515	473	469	463	471	513	509	544
	CDD18.3	3065	295	273	302	278	257	223	211	205	221	255	259	286
	CDH23.3	30647	3347	3253	3652	3213	2619	1902	1534	1429	1751	2343	2554	3049
CDH26.7	7874	1057	1151	1324	1058	656	282	143	121	223	447	571	842	
Wind	WSAvg	4.3	4.4	4.3	4.1	3.7	3.5	3.8	4.2	4.3	4.6	4.8	4.7	4.6
Precipitation	PrecAvg	1392	53	83	114	197	259	197	163	101	69	60	52	32
	PrecMax	2359	301	279	374	475	650	452	435	198	192	221	317	133
	PrecMin	729	1	6	4	19	44	67	42	33	3	0	1	0
	PrecStd	402	57	73	87	119	151	80	68	37	43	61	63	30
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.1	32.9	33.0	32.2	31.2	29.9	29.1	29.0	29.8	30.6	30.9	31.2
		MCWB	26.9	27.0	27.1	26.4	26.3	25.3	24.6	24.4	25.4	25.7	26.5	26.3
	2%	DB	31.2	32.0	32.1	31.2	30.2	29.1	28.2	28.2	29.0	30.0	30.1	31.0
		MCWB	26.4	26.6	26.5	26.1	25.5	24.5	23.6	23.4	24.5	25.2	25.4	26.2
	5%	DB	30.9	31.2	31.2	30.9	30.0	28.8	28.0	28.0	28.2	29.2	29.8	30.2
		MCWB	26.2	26.2	26.2	26.0	25.3	24.3	23.4	23.2	23.6	24.5	25.0	25.4
	10%	DB	30.1	30.8	30.8	30.1	29.2	28.1	27.2	27.2	28.0	28.9	29.1	29.9
		MCWB	25.2	25.9	26.0	25.4	24.7	23.7	22.9	22.8	23.4	24.3	24.5	25.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.4	27.5	27.7	27.3	26.8	26.2	25.6	25.4	25.8	26.4	26.7	26.9
		MCDB	31.0	31.8	31.6	30.2	29.6	28.4	27.8	27.5	28.5	29.7	29.9	30.5
	2%	WB	26.7	26.9	27.1	26.8	26.4	25.6	25.1	24.8	25.2	25.7	26.1	26.5
		MCDB	30.2	30.9	30.9	29.9	29.3	27.9	27.3	27.0	27.9	28.8	29.0	30.0
	5%	WB	26.5	26.6	26.7	26.5	26.0	25.1	24.6	24.3	24.7	25.2	25.6	26.1
		MCDB	29.8	30.4	30.2	29.6	28.9	27.5	26.8	26.5	27.3	28.2	28.5	29.4
	10%	WB	26.1	26.3	26.4	26.2	25.6	24.7	24.1	23.9	24.3	24.8	25.2	25.6
		MCDB	29.2	29.8	29.8	29.2	28.4	27.1	26.4	26.1	26.9	27.6	28.0	28.8
Mean Daily Temperature Range	5% DB	MDBR	4.9	4.9	5.2	5.0	5.2	4.9	4.8	4.8	4.5	4.5	4.6	4.8
		MCDBR	5.5	5.5	5.8	5.6	5.7	5.4	5.2	5.3	4.8	4.8	4.9	5.0
		MCWBR	2.6	2.4	2.7	2.5	2.8	2.8	2.7	2.7	2.4	2.3	2.4	2.4
	5% WB	MCDBR	5.4	5.4	5.5	5.2	5.4	5.0	5.0	4.8	4.8	4.8	4.8	4.9
		MCWBR	2.8	2.7	2.9	2.8	3.0	3.0	3.0	3.0	2.6	2.6	2.6	2.6
Clear-Sky Solar Irradiance	taub	0.390	0.386	0.383	0.379	0.378	0.369	0.370	0.377	0.411	0.412	0.403	0.403	
	taud	2.508	2.534	2.548	2.572	2.575	2.582	2.568	2.514	2.397	2.414	2.450	2.451	
	Ebn at Noon	952	950	935	909	880	873	879	898	896	914	932	937	
	Edh at Noon	114	111	107	99	94	91	94	104	122	124	120	120	
All-Sky Solar Radiation	RadAvg	6.58	6.42	6.26	5.58	4.86	4.59	4.74	5.29	5.96	6.43	6.71	6.78	
	RadStd	0.47	0.30	0.29	0.33	0.34	0.26	0.19	0.25	0.27	0.25	0.41	0.28	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page